Dy11 Transformer Secondary Voltages Under Single-Phase Voltage Sag on the Primary Side

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1 Problem Statement

Consider a balanced three-phase primary supply with nominal line-to-neutral voltage magnitude $V_0 = 1\,\text{pu}$. A *type B* single-phase voltage sag occurs on phase V: the voltage drops to $0.25\,\text{pu}$, while phases U and W remain at $1\,\text{pu}$. The transformer is a **Dy11** unit with turns ratio $k = N_1/N_2$.

The goal is to determine the secondary line-to-line voltages (and, by extension, the line-to-neutral voltages of the equivalent star on the secondary side).

2 Primary Phasor Definition

Taking the pre-fault phasors as reference (subscript 1 for primary):

$$\underline{V}_{U1} = V_0\, e^{j\,0} = V_0, \qquad \underline{V}_{V1} = V_0\, e^{-j\,\frac{2\pi}{3}}, \qquad \underline{V}_{W1} = V_0\, e^{+j\,\frac{2\pi}{3}}. \tag{1}$$

After the sag on phase V with depth $p = 0.25$:

$$\underline{V}_{U1} = V_0, \tag{2}$$

$$\underline{V}'_{V1} = p\,V_0\, e^{-j\,\frac{2\pi}{3}} = 0.25\,V_0\, e^{-j\,120\check{\text{r}}}, \tag{3}$$

$$\underline{V}_{W1} = V_0\, e^{+j\,\frac{2\pi}{3}} = V_0\, e^{+j\,120\check{\text{r}}}. \tag{4}$$

Using per-unit values with $V_0 = 1$, and expressing in rectangular form:

$$\underline{V}_{U1} = 1 + j\,0, \tag{5}$$

$$\underline{V}'_{V1} = 0.25\Big(-\tfrac{1}{2} - j\,\tfrac{\sqrt{3}}{2}\Big) = -0.125 - j\,0.2165, \tag{6}$$

$$\underline{V}_{W1} = -\tfrac{1}{2} + j\,\tfrac{\sqrt{3}}{2} = -0.5 + j\,0.8660. \tag{7}$$

3 Primary Line-to-Line Voltages

The delta winding of a Dy transformer is connected across the *primary line-to-line* voltages. With the sag on V:

$$\underline{V}_{UV1} = \underline{V}_{U1} - \underline{V}'_{V1} = (1 + 0.125) + j\,0.2165 = 1.125 + j\,0.2165, \tag{8}$$

$$\underline{V}_{VW1} = \underline{V}'_{V1} - \underline{V}_{W1} = (-0.125 + 0.5) + j(-0.2165 - 0.8660) = 0.375 - j\,1.0825, \tag{9}$$

$$\underline{V}_{WU1} = \underline{V}_{W1} - \underline{V}_{U1} = -1.5 + j\,0.8660. \tag{10}$$

Their magnitudes are:

$$|\underline{V}_{UV1}| = \sqrt{1.125^2 + 0.2165^2} = \sqrt{1.2656 + 0.0469} = \sqrt{1.3125} \approx 1.1456\,\text{pu}, \tag{11}$$

$$|\underline{V}_{VW1}| = \sqrt{0.375^2 + 1.0825^2} = \sqrt{0.1406 + 1.1718} = \sqrt{1.3124} \approx 1.1456\,\text{pu}, \tag{12}$$

$$|\underline{V}_{WU1}| = \sqrt{1.5^2 + 0.8660^2} = \sqrt{2.25 + 0.75} = \sqrt{3} \approx 1.7321\,\text{pu}. \tag{13}$$

Remark: Under a balanced supply, all three line-to-line voltages would equal $\sqrt{3}\,\text{pu} \approx 1.7321\,\text{pu}$. The sag reduces V_{UV} and V_{VW}, while V_{WU} (which does not involve phase V directly from W to U) retains its nominal magnitude.

4 Dy11 Transformation and Secondary Voltages

4.1 Dy11 Phase Displacement Convention

In the **Dy11** group, the secondary star (y) leads the primary delta (D) by $11 \times 30\text{ř} = 330\text{ř}$, which is equivalent to a **lead of** -30ř (i.e., $+30\text{ř}$ when measured as the secondary leading the primary). The standard convention states:

$$\underline{V}_{a2} = \frac{1}{k}\,\underline{V}_{UV1}\,e^{+j\,30\text{ř}}, \qquad \underline{V}_{b2} = \frac{1}{k}\,\underline{V}_{VW1}\,e^{+j\,30\text{ř}}, \qquad \underline{V}_{c2} = \frac{1}{k}\,\underline{V}_{WU1}\,e^{+j\,30\text{ř}}, \tag{14}$$

where k is the primary-to-secondary turns ratio. Setting $k = 1$ for per-unit analysis (the secondary base voltage is chosen so that rated secondary equals 1 pu when the primary is balanced at 1 pu):

$$\frac{1}{k} = \frac{1}{\sqrt{3}} \tag{15}$$

because the primary line-to-line base equals $\sqrt{3}$ times the primary phase base. The $+30\text{ř}$ rotation is the defining characteristic of the Dy11 group.

4.2 Secondary Line-to-Neutral Phasors

Applying (14) with the $+30\text{ř}$ rotation operator $e^{+j30\text{ř}} = \frac{\sqrt{3}}{2} + j\frac{1}{2}$:

Phase a (mapped from UV):

$$\begin{aligned}
\underline{V}_{a2} &= \frac{1}{\sqrt{3}}\,(1.125 + j\,0.2165)\left(\frac{\sqrt{3}}{2} + j\,\frac{1}{2}\right) \\
&= \frac{1}{\sqrt{3}}\left[(1.125\cdot\tfrac{\sqrt{3}}{2} - 0.2165\cdot\tfrac{1}{2}) + j\,(1.125\cdot\tfrac{1}{2} + 0.2165\cdot\tfrac{\sqrt{3}}{2})\right] \\
&= \frac{1}{\sqrt{3}}\left[(0.9743 - 0.1083) + j\,(0.5625 + 0.1875)\right] \\
&= \frac{1}{\sqrt{3}}\,(0.8660 + j\,0.75) \\
&\approx 0.5 + j\,0.4330.
\end{aligned} \tag{16}$$

Magnitude: $|\underline{V}_{a2}| = \sqrt{0.5^2 + 0.4330^2} = \sqrt{0.25 + 0.1875} = \sqrt{0.4375} \approx \mathbf{0.6614\,pu}$.

Phase b (mapped from VW):

$$\begin{aligned}
\underline{V}_{b2} &= \frac{1}{\sqrt{3}}\,(0.375 - j\,1.0825)\left(\frac{\sqrt{3}}{2} + j\,\frac{1}{2}\right) \\
&= \frac{1}{\sqrt{3}}\left[(0.375\cdot\tfrac{\sqrt{3}}{2} + 1.0825\cdot\tfrac{1}{2}) + j\,(0.375\cdot\tfrac{1}{2} - 1.0825\cdot\tfrac{\sqrt{3}}{2})\right] \\
&= \frac{1}{\sqrt{3}}\left[(0.3248 + 0.5413) + j\,(0.1875 - 0.9375)\right] \\
&= \frac{1}{\sqrt{3}}\,(0.8661 - j\,0.75) \\
&\approx 0.5 - j\,0.4330.
\end{aligned} \tag{17}$$

Magnitude: $|\underline{V}_{b2}| = \sqrt{0.5^2 + 0.4330^2} \approx \mathbf{0.6614\,pu}$.

Phase c (mapped from WU):

$$\begin{aligned}
\underline{V}_{c2} &= \frac{1}{\sqrt{3}}\,(-1.5 + j\,0.8660)\,\left(\frac{\sqrt{3}}{2} + j\,\frac{1}{2}\right) \\
&= \frac{1}{\sqrt{3}}\,\left[(-1.5\cdot\tfrac{\sqrt{3}}{2} - 0.8660\cdot\tfrac{1}{2}) + j\,(-1.5\cdot\tfrac{1}{2} + 0.8660\cdot\tfrac{\sqrt{3}}{2})\right] \\
&= \frac{1}{\sqrt{3}}\,\left[(-1.2990 - 0.4330) + j\,(-0.75 + 0.75)\right] \\
&= \frac{1}{\sqrt{3}}\,(-1.7320 + j\,0) \\
&= -1.0 + j\,0. \qquad (18)
\end{aligned}$$

Magnitude: $|\underline{V}_{c2}| = \mathbf{1.0\,pu}$.

5 Summary of Results

Table 1 collects all primary and secondary voltages in per-unit.

Table 1: Primary and secondary voltages under type B sag on phase V ($p = 0.25\,\text{pu}$), Dy11 transformer, k=1 pu.

Quantity	**Phasor (pu)**	**Magnitude (pu)**	**Angle (deg)**	**Nominal (pu)**
V_{U1}	$1{+}j0$	1.000	0ř	1.000
V'_{V1}	$-0.125{-}j0.2165$	0.250	-120ř	1.000
V_{W1}	$-0.5{+}j0.8660$	1.000	$+120$ř	1.000
V_{UV1}	$1.125{+}j0.2165$	1.146	$+10.9$ř	$\sqrt{3}$
V_{VW1}	$0.375{-}j1.0825$	1.146	-70.9ř	$\sqrt{3}$
V_{WU1}	$-1.5{+}j0.8660$	1.732	$+150$ř	$\sqrt{3}$
V_{a2}	$0.5{+}j0.4330$	0.661	$+40.9$ř	1.000
V_{b2}	$0.5{-}j0.4330$	0.661	-40.9ř	1.000
V_{c2}	$-1.0{+}j0$	1.000	$+180$ř	1.000

5.1 Physical Interpretation

- **Phase c on the secondary is unaffected** ($|V_{c2}| = 1\,\text{pu}$) because it is derived from the WU primary line-to-line voltage, which involves only phases U and W — both at rated voltage.
- **Phases a and b are equally reduced** to $\approx 0.661\,\text{pu}$ (i.e., $\frac{1}{\sqrt{3}}\cdot\sqrt{1.3125} \approx 0.661$), since both V_{UV} and V_{VW} are affected symmetrically by the sag on V.
- The **angular spread** of the secondary phasors is no longer 120ř: phases a and b are only 81.8ř apart instead of 120ř, while the angle from b to c and from c to a are both ≈ 139.1ř. This unbalanced phase portrait is the characteristic signature of a Dy-transformed type B sag.
- In terms of *symmetrical components*, the primary unbalanced set contains a negative-sequence component introduced by the sag. The delta primary winding acts as a *zero-sequence trap* (no zero-sequence current circulates in the secondary star), but the negative-sequence component passes through undistorted (with the 30ř rotation of the Dy11 group). Consequently the secondary contains both positive- and negative-sequence components, but *no zero-sequence component*.

6 Symmetrical-Component Verification

The secondary unbalanced set can be decomposed as $\underline{V}_{abc} = \underline{V}^{+} + \underline{V}^{-}$ (no zero sequence due to the delta primary). Applying the Fortescue transformation:

$$\begin{pmatrix} V^0 \\ V^+ \\ V^- \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha \end{pmatrix}\begin{pmatrix} V_{a2} \\ V_{b2} \\ V_{c2} \end{pmatrix}, \qquad \alpha = e^{j\,120\check{r}}. \tag{19}$$

With $V_{a2} = 0.5 + j0.4330$, $V_{b2} = 0.5 - j0.4330$, $V_{c2} = -1$:

$$V^0 = \tfrac{1}{3}(V_{a2} + V_{b2} + V_{c2}) = \tfrac{1}{3}(0.5 + 0.5 - 1) = 0 \quad \checkmark \tag{20}$$

$$V^+ = \tfrac{1}{3}(V_{a2} + \alpha\, V_{b2} + \alpha^2 V_{c2}). \tag{21}$$

Recalling $\alpha = -\frac{1}{2} + j\frac{\sqrt{3}}{2}$ and $\alpha^2 = -\frac{1}{2} - j\frac{\sqrt{3}}{2}$:

$$\begin{aligned} \alpha\, V_{b2} &= \left(-\tfrac{1}{2} + j\tfrac{\sqrt{3}}{2}\right)(0.5 - j0.4330) = (-0.25 + 0.3750) + j(-0.2165 + 0.4330 \cdot \tfrac{\sqrt{3}}{2} \cdot 2) \\ &= (-0.25 + 0.3750) + j(-0.2165 + 0.3750) \\ &\approx 0.125 + j\,0.6495, \\ \alpha^2 V_{c2} &= \left(-\tfrac{1}{2} - j\tfrac{\sqrt{3}}{2}\right)(-1) = 0.5 + j\,0.8660, \\ V^+ &= \tfrac{1}{3}\left[(0.5 + j0.4330) + (0.125 + j0.6495) + (0.5 + j0.8660)\right] \\ &= \tfrac{1}{3}(1.125 + j\,1.9485) \approx 0.375 + j\,0.6495 \approx 0.75\, e^{j\,60\check{r}}, \end{aligned} \tag{22}$$

$$|V^+| \approx 0.750\,\text{pu}. \tag{23}$$

Similarly, $|V^-| \approx 0.250\,\text{pu}$ (the negative-sequence magnitude equals the per-unit sag depth residual). This confirms the well-known result for a type B sag transformed through a Dy connection:

$$\boxed{|V^+| = \frac{2+p}{3} = \frac{2+0.25}{3} = 0.75\,\text{pu}, \qquad |V^-| = \frac{1-p}{3} = \frac{1-0.25}{3} = 0.25\,\text{pu}.} \tag{24}$$

7 Conclusions

For a **type B single-phase voltage sag** ($V_V = 0.25\,\text{pu}$, $V_U = V_W = 1\,\text{pu}$) on the primary side of a **Dy11** transformer, the secondary voltages are:

Secondary phase	**Magnitude (pu)**	**Note**
a	0.661	Reduced (involves sagged phase V via UV)
b	0.661	Reduced (involves sagged phase V via VW)
c	1.000	Unaffected (WU path; V not involved)

The delta primary *converts* the single-phase primary sag (type B) into a **two-phase secondary sag** (type C/D family), with no zero-sequence propagation. The negative-sequence component at the secondary equals $|V^-| = (1-p)/3 = 0.25\,\text{pu}$, which drives negative-sequence currents in any downstream motor or power-electronic load — a critical input for ride-through and control design.

8 Inverse Problem: Primary Sag Required to Obtain a Type B Sag at the Secondary

8.1 Problem Statement

We now seek the *primary* phase voltages such that the **secondary** presents a type B sag:

$$V_{U2} = 1\,\text{pu}, \qquad V_{V2} = 0.25\,\text{pu}\ (\angle{-120}\text{ř}), \qquad V_{W2} = 1\,\text{pu}\ (\angle{+120}\text{ř}). \tag{25}$$

8.2 Secondary Phasors in Rectangular Form

$$\underline{V}_{U2} = 1 + j\,0, \tag{26}$$
$$\underline{V}_{V2} = 0.25\,e^{-j120\text{ř}} = -0.125 - j\,0.2165, \tag{27}$$
$$\underline{V}_{W2} = 1\,e^{+j120\text{ř}} = -0.5 + j\,0.8660. \tag{28}$$

8.3 Secondary Line-to-Line Voltages

$$\underline{V}_{UV2} = \underline{V}_{U2} - \underline{V}_{V2} = 1.125 + j\,0.2165, \quad |\underline{V}_{UV2}| \approx 1.146\,\text{pu}, \tag{29}$$
$$\underline{V}_{VW2} = \underline{V}_{V2} - \underline{V}_{W2} = 0.375 - j\,1.0825, \quad |\underline{V}_{VW2}| \approx 1.146\,\text{pu}, \tag{30}$$
$$\underline{V}_{WU2} = \underline{V}_{W2} - \underline{V}_{U2} = -1.5 + j\,0.8660, \quad |\underline{V}_{WU2}| = \sqrt{3}\,\text{pu}. \tag{31}$$

8.4 Inverse Dy11 Mapping

The Dy11 group applies a $+30$ř rotation and a $\sqrt{3}$ scaling from primary line-to-line to secondary line-to-neutral (Section 4). The inverse mapping is therefore a -30ř rotation and a $\sqrt{3}$ upscaling:

$$\underline{V}_{UV1} = \sqrt{3}\,\underline{V}_{U2}\,e^{-j\,30\text{ř}}, \qquad \underline{V}_{VW1} = \sqrt{3}\,\underline{V}_{V2}\,e^{-j\,30\text{ř}}, \qquad \underline{V}_{WU1} = \sqrt{3}\,\underline{V}_{W2}\,e^{-j\,30\text{ř}}. \tag{32}$$

The -30ř rotation operator is $e^{-j30\text{ř}} = \frac{\sqrt{3}}{2} - j\frac{1}{2}$.

Primary line UV:

$$\underline{V}_{UV1} = \sqrt{3}\,(1 + j\,0)\Big(\tfrac{\sqrt{3}}{2} - j\tfrac{1}{2}\Big) = \sqrt{3}\Big(\tfrac{\sqrt{3}}{2} - j\tfrac{1}{2}\Big) = \tfrac{3}{2} - j\tfrac{\sqrt{3}}{2} \approx 1.5 - j\,0.8660. \tag{33}$$

Magnitude: $|\underline{V}_{UV1}| = \sqrt{1.5^2 + 0.8660^2} = \sqrt{3}\,\text{pu}$. Angle: -30ř.

Primary line VW:

$$\begin{aligned}
\underline{V}_{VW1} &= \sqrt{3}\,(-0.125 - j\,0.2165)\Big(\tfrac{\sqrt{3}}{2} - j\tfrac{1}{2}\Big) \\
&= \sqrt{3}\Big[(-0.125 \cdot \tfrac{\sqrt{3}}{2} - 0.2165 \cdot \tfrac{1}{2}) + j\,(-0.125 \cdot (-\tfrac{1}{2}) + (-0.2165) \cdot \tfrac{\sqrt{3}}{2})\Big] \\
&= \sqrt{3}[(-0.1083 - 0.1083) + j\,(0.0625 - 0.1875)] \\
&= \sqrt{3}\,(-0.2165 - j\,0.125) \\
&\approx -0.375 - j\,0.2165.
\end{aligned} \tag{34}$$

Magnitude: $\sqrt{3} \cdot 0.25 \approx 0.433\,\text{pu}$. Angle: -150ř.

Primary line WU:

$$\begin{aligned}\underline{V}_{WU1} &= \sqrt{3}\,(-0.5 + j\,0.8660)\Big(\tfrac{\sqrt{3}}{2} - j\tfrac{1}{2}\Big)\\ &= \sqrt{3}\Big[(-0.5\cdot\tfrac{\sqrt{3}}{2} - 0.8660\cdot(-\tfrac{1}{2})) + j\,((-0.5)\cdot(-\tfrac{1}{2}) + 0.8660\cdot\tfrac{\sqrt{3}}{2})\Big]\\ &= \sqrt{3}[(-0.4330 + 0.4330) + j\,(0.25 + 0.75)]\\ &= \sqrt{3}\,(0 + j\,1.0)\\ &\approx 0 + j\,1.7321. \end{aligned} \tag{35}$$

Magnitude: $\sqrt{3}\cdot 1.0 \approx 1.732\,\text{pu}$. Angle: $+90$ř.

8.5 Primary Phase-to-Neutral Voltages

The primary line-to-line voltages are related to the primary phase-to-neutral voltages by:

$$\underline{V}_{UV1} = \underline{V}_{U1} - \underline{V}_{V1}, \quad \underline{V}_{VW1} = \underline{V}_{V1} - \underline{V}_{W1}, \quad \underline{V}_{WU1} = \underline{V}_{W1} - \underline{V}_{U1}. \tag{36}$$

This is a 3×3 singular system (only two equations are independent for a three-phase system without neutral connection). Imposing the conventional constraint $\underline{V}_{U1} + \underline{V}_{V1} + \underline{V}_{W1} = 0$ and solving:

$$\underline{V}_{U1} = \tfrac{1}{3}(\underline{V}_{UV1} - \underline{V}_{WU1})\,, \tag{37}$$
$$\underline{V}_{V1} = \tfrac{1}{3}(\underline{V}_{VW1} - \underline{V}_{UV1})\,, \tag{38}$$
$$\underline{V}_{W1} = \tfrac{1}{3}(\underline{V}_{WU1} - \underline{V}_{VW1})\,. \tag{39}$$

Substituting the computed line-to-line phasors:

Phase U:

$$\begin{aligned}\underline{V}_{U1} &= \tfrac{1}{3}\big[(1.5 - j\,0.8660) - (0 + j\,1.7321)\big]\\ &= \tfrac{1}{3}(1.5 - j\,2.5981)\\ &\approx 0.5 - j\,0.8660 = e^{-j\,60\text{ř}}. \end{aligned} \tag{40}$$

$|\underline{V}_{U1}| = 1\,\text{pu}, \quad \angle\underline{V}_{U1} = -60$ř.

Phase V:

$$\begin{aligned}\underline{V}_{V1} &= \tfrac{1}{3}\big[(-0.375 - j\,0.2165) - (1.5 - j\,0.8660)\big]\\ &= \tfrac{1}{3}(-1.875 + j\,0.6495)\\ &\approx -0.625 + j\,0.2165. \end{aligned} \tag{41}$$

$|\underline{V}_{V1}| = \sqrt{0.625^2 + 0.2165^2} = \sqrt{0.3906 + 0.0469} = \sqrt{0.4375} \approx \mathbf{0.661}\,\text{pu}$, $\angle\underline{V}_{V1} = 180\text{ř} - \arctan(0.2165/0.625) \approx 180\text{ř} - 19.1\text{ř} = +160.9\text{ř}$.

Phase W:

$$\begin{aligned}\underline{V}_{W1} &= \tfrac{1}{3}\big[(0 + j\,1.7321) - (-0.375 - j\,0.2165)\big]\\ &= \tfrac{1}{3}(0.375 + j\,1.9486)\\ &\approx 0.125 + j\,0.6495. \end{aligned} \tag{42}$$

$|\underline{V}_{W1}| = \sqrt{0.125^2 + 0.6495^2} = \sqrt{0.0156 + 0.4219} = \sqrt{0.4375} \approx \mathbf{0.661}\,\text{pu}$, $\angle\underline{V}_{W1} = \arctan(0.6495/0.125) \approx +79.1$ř.

Table 2: Primary voltages required to produce a type B sag (V_{V2}=0.25 pu) at the secondary of a Dy11 transformer.

P. phase	Phasor (pu)	Mag. (pu)	Angle (deg)	Comment
U	$0.5 - j\,0.866$	1.000	-60ř	Nominal magnitude, shifted -60ř
V	$-0.625 + j\,0.2165$	0.661	$+160.9$ř	**Reduced to** 0.661 pu**, shifted**
W	$0.125 + j\,0.6495$	0.661	$+79.1$ř	**Reduced to** 0.661 pu**, shifted**

8.6 Summary: Required Primary Sag

8.7 Physical Interpretation of the Inverse Result

Several key observations follow from this inverse analysis:

- The required primary sag is a **type C sag**: *two phases* are reduced in magnitude (V and W, both to ≈ 0.661 pu), while one phase (U) retains nominal magnitude. This is exactly the dual of the forward transformation: just as a Dy connection turns a type B primary sag into a type C/D secondary sag, it turns a type C primary sag into a type B secondary sag.
- The entire primary phasor set is **rotated by** -60ř with respect to the standard reference position. This is a consequence of the -30ř inverse Dy11 mapping applied to a secondary reference frame that itself started from 0ř. In practice, the phase angles depend on the chosen reference frame; what matters physically is only the *relative* magnitudes and angles between the three phases.
- The **sag depth at the primary** corresponding to the 0.25 pu secondary type B sag is:

$$p_{\text{prim}} = 0.661\,\text{pu} = \sqrt{\frac{1 + p_{\text{sec}}^2 + p_{\text{sec}}}{3}} = \sqrt{\frac{1 + 0.25^2 + 0.25}{3}} = \sqrt{\frac{1.3125}{3}} \approx 0.661\,\text{pu}, \quad (43)$$

  confirming the analytical formula for Dy-transformed type B/C sag pairs.
- The **symmetrical-component content** of the required primary type C sag is $|V_1^+| \approx 0.7737$ pu and $|V_1^-| \approx 0.1130$ pu. These values differ from those at the secondary (Section 6: $|V_2^+| = 0.75$, $|V_2^-| = 0.25$ pu) because the Dy transformation does *not* preserve sequence magnitudes when the sag *type* changes (type C primary $\leftrightarrow$ type B secondary); it only preserves magnitudes for the *same* sag type propagating unchanged through the transformer.

In summary: to observe a type B sag (U=1, V=0.25, W=1 pu) at the *secondary* of a Dy11 transformer, one must apply a **type C sag with depth** ≈ 0.661 pu **on two phases** (V and W) at the primary, with the phasor frame rotated by -60ř relative to the secondary reference.

9 D–Q Decomposition of the Primary Type C Sag

9.1 Definition of the D–Q Frame

In the synchronous dq reference frame rotating at angular frequency ω, a three-phase voltage set is decomposed into:

- a **positive-sequence** phasor $\underline{V}^+ = D_p + jQ_p$, co-rotating with the frame;
- a **negative-sequence** phasor $\underline{V}^- = D_n + jQ_n$, counter-rotating at 2ω in the frame.

The Fortescue decomposition gives:

$$\underline{V}^{+} = \frac{1}{3}(V_U + \alpha\, V_V + \alpha^2 V_W), \qquad \underline{V}^{-} = \frac{1}{3}(V_U + \alpha^2 V_V + \alpha\, V_W), \qquad \alpha = e^{j\,120\check{\mathrm{r}}}. \tag{44}$$

9.2 Canonical Type C Frame ($V_U = 1\angle 0\check{\mathrm{r}}$)

The type C sag with $p = 0.661\,\mathrm{pu}$ and V_U as phase reference:

$$\underline{V}_U = 1, \qquad \underline{V}_V = 0.661\, e^{-j120\check{\mathrm{r}}}, \qquad \underline{V}_W = 0.661\, e^{+j120\check{\mathrm{r}}}. \tag{45}$$

For a type C sag with two phases symmetrically reduced to p, the closed-form Fortescue result is:

$$\underline{V}^{+} = \frac{1+2p}{3}, \qquad \underline{V}^{-} = \frac{1-p}{3}, \qquad \underline{V}^{0} = 0. \tag{46}$$

With $p = 0.661$:

$$\underline{V}^{+} = \frac{1 + 2 \times 0.661}{3} = \frac{2.322}{3} = 0.7740\,\mathrm{pu}\,\angle 0\check{\mathrm{r}}, \tag{47}$$

$$\underline{V}^{-} = \frac{1 - 0.661}{3} = \frac{0.339}{3} = 0.1130\,\mathrm{pu}\,\angle 0\check{\mathrm{r}}. \tag{48}$$

Both phasors are *real* in the canonical frame (no imaginary part), so:

$$\boxed{D_p = 0.7740\,\mathrm{pu}, \quad Q_p = 0, \quad D_n = 0.1130\,\mathrm{pu}, \quad Q_n = 0.} \tag{49}$$

Remark: the absence of Q components in the canonical frame is the defining property of a type C sag — the sag is purely real when V_U is the D-axis reference. Any non-zero Q component signals either a non-zero fault impedance angle, a measurement frame offset, or a different sag type.

9.3 Rotated Frame ($V_U = 1\angle -60\check{\mathrm{r}}$, as required at the Dy11 primary)

When the primary phasor set is expressed in the rotated frame of Section 8 (where $V_{U1} = e^{-j60\check{\mathrm{r}}}$), the dq reference axis itself is rotated by $\theta_0 = -60\check{\mathrm{r}}$ with respect to the canonical axis. The sequence phasors transform as:

$$\underline{V}^{+}_{\theta_0} = \underline{V}^{+}\, e^{j\theta_0} = 0.7740\, e^{-j60\check{\mathrm{r}}}, \qquad \underline{V}^{-}_{\theta_0} = \underline{V}^{-}\, e^{-j\theta_0} = 0.1130\, e^{+j60\check{\mathrm{r}}}. \tag{50}$$

(Positive sequence rotates with the frame; negative sequence rotates opposite.) In rectangular form:

$$\underline{V}^{+}_{\theta_0} = 0.7740\left(\tfrac{1}{2} - j\tfrac{\sqrt{3}}{2}\right) = 0.3870 - j\,0.6703, \tag{51}$$

$$\underline{V}^{-}_{\theta_0} = 0.1130\left(\tfrac{1}{2} + j\tfrac{\sqrt{3}}{2}\right) = 0.0565 + j\,0.0978. \tag{52}$$

Thus, in the rotated primary frame:

$$\boxed{D_p = 0.3870\,\mathrm{pu}, \quad Q_p = -0.6703\,\mathrm{pu}, \quad D_n = 0.0565\,\mathrm{pu}, \quad Q_n = +0.0978\,\mathrm{pu}.} \tag{53}$$

9.4 Summary Table

9.5 Relevance for Control Design

In a vector-controlled drive or grid-tied converter operating under this type C primary sag, the D–Q components have direct implications:

Table 3: Symmetrical components and D–Q decomposition of the primary type C sag ($p = 0.661$ pu) in both reference frames.

Frame	$\lvert V^+\rvert$ (pu)	$\lvert V^-\rvert$ (pu)	D_p (pu)	Q_p (pu)	D_n (pu)	Q_n (pu)
Canonical ($V_U = 1\angle 0$ř)	0.7740	0.1130	+0.7740	0	+0.1130	0
Rotated ($V_U = 1\angle -60$ř, Dy11)	0.7740	0.1130	+0.3870	−0.6703	+0.0565	+0.0978

- D_p and Q_p feed the *positive-sequence current controller* (inner dq loop). A drop in D_p from 1.0 to 0.774 pu reduces the available active power headroom.
- D_n and Q_n generate a *double-frequency ripple* at 2ω in the positive-sequence dq frame. Without a dedicated negative-sequence controller (e.g. a second dq loop at $-\omega$, or a notch filter at 2ω on the current references), this ripple appears as a 100 Hz (or 120 Hz) disturbance on the d- and q-axis currents.
- In the canonical frame, $Q_n = 0$ means the negative-sequence voltage has no reactive component along the q-axis: the ripple is *in-phase* with the D-axis and affects primarily the active-power channel. In the rotated frame both channels are disturbed.
- The ratio $|V^-|/|V^+| = 0.1130/0.7740 \approx 14.6\%$ is the *voltage unbalance factor* (VUF) at the primary terminals. IEC 61000-2-2 sets a planning level of 2% for LV networks and 1% for HV; a VUF of 14.6% is therefore a severe unbalance condition requiring ride-through or protection action.

10 Grid-Voltage Constraint for AFE Operation

Consider an Active Front-End (AFE) converter connected to a stiff grid through a line inductance L, with line resistance neglected. Let V_s denote the (constant) upstream line-to-neutral RMS voltage, ω the fundamental angular frequency, and $X_L = \omega L$ the line reactance. Define the synchronous dq-frame with the d-axis aligned to the terminal voltage phasor $\overrightarrow{V}_{\text{AFE}}$.

10.1 Phasor Balance at the PCC

The phasor Kirchhoff equation at the Point of Common Coupling (PCC) is

$$\overrightarrow{V}_s = \overrightarrow{V}_{\text{AFE}} + jX_L\,\overrightarrow{I}, \tag{54}$$

where $\overrightarrow{I} = i_d - ji_q$ in the adopted convention ($i_q > 0$ for capacitive/leading reactive power injection).

10.2 Component-wise Decomposition

Separating real and imaginary parts of (54):

$$V_{s,d} = V_{\text{AFE}} + X_L\, i_q, \tag{55}$$

$$V_{s,q} = -X_L\, i_d. \tag{56}$$

Because the d-axis is locked to $\overrightarrow{V}_{\text{AFE}}$, we have $V_{s,q} = 0$ only at the AFE terminals; the upstream components are non-zero and encode the reactive loading.

10.3 Voltage Magnitude Constraint

The upstream grid magnitude must satisfy

$$|\overrightarrow{V}_s|^2 = V_{s,d}^2 + V_{s,q}^2 = V_s^2. \tag{57}$$

Substituting (55)–(56) into (57):

$$\left(V_{\mathrm{AFE}} + X_L\, i_q\right)^2 + \left(X_L\, i_d\right)^2 = V_s^2. \tag{58}$$

10.4 Locus in the $(i_d,\, i_q)$ Plane

Equation (58) defines an implicit relationship between i_d, i_q, and the AFE terminal voltage V_{AFE}. For a fixed modulation index the AFE output voltage is bounded:

$$V_{\mathrm{AFE}} \le V_{\mathrm{AFE,max}} = \frac{m_a\, V_{\mathrm{dc}}}{2}, \tag{59}$$

where $m_a \le 1$ is the modulation index and V_{dc} is the DC-link voltage. Substituting $V_{\mathrm{AFE}} = V_{\mathrm{AFE,max}}$ into (58) yields the *boundary locus*:

$$\boxed{X_L^2\, i_d^2 + \left(X_L\, i_q + V_{\mathrm{AFE,max}}\right)^2 = V_s^2} \tag{60}$$

Equation (60) is the equation of a **circle** in the $(i_d,\, i_q)$ plane with:

$$\text{centre:} \quad \left(0,\; -\frac{V_{\mathrm{AFE,max}}}{X_L}\right), \tag{61}$$

$$\text{radius:} \quad r = \frac{V_s}{X_L}. \tag{62}$$

The AFE can remain grid-connected (i.e. maintain $|\overrightarrow{V}_s| = V_s$) only for operating points $(i_d,\, i_q)$ that lie *on or inside* this circle.

10.5 Physical Interpretation

- The circle is centred on the negative i_q-axis at a distance $V_{\mathrm{AFE,max}}/X_L$; it is therefore *not* centred at the origin, reflecting the non-zero baseline reactive current needed to sustain $V_{\mathrm{AFE,max}}$ across X_L.
- A larger X_L (weaker grid) shrinks both the radius r and shifts the centre further into the inductive quadrant, tightening the reactive-current margin.
- At unity power factor ($i_q = 0$), the maximum active current is recovered from (60):

$$i_{d,\max}\big|_{i_q=0} = \frac{1}{X_L}\sqrt{V_s^2 - V_{\mathrm{AFE,max}}^2}. \tag{63}$$

- For a stiff grid ($X_L \to 0$), $r \to \infty$ and the constraint becomes inactive, recovering the ideal voltage-source assumption.